

Tutorial 1

1.

The velocity components in a two-dimensional flow field for an incompressible fluid are

$$u = \frac{y^3}{3} - x^2y + 2x \text{ and } v = -\frac{x^3}{3} + xy^2 - 2y$$

- i) Show that these functions represent an irrotational flow.
- ii) Obtain an expression for stream function ψ .

Solution:

$$\text{i) } \frac{\partial u}{\partial x} = 2 - 2xy; \frac{\partial v}{\partial y} = 2xy - 2$$

For a two-dimensional flow of incompressible fluid, the continuity is expressed as

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

By substituting for the two terms, we have

$$2 - 2xy + 2xy - 2 = 0$$

Thus, continuity equation is satisfied. Hence fluid flow is possible.

Further,

$$\frac{\partial v}{\partial x} = y^2 - x^2; \frac{\partial u}{\partial y} = y^2 - x^2$$

On substituting the above in conditions for irrotationality, we get $\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = y^2 - x^2 - y^2 + x^2 = 0$ Hence, the flow is irrotational.

ii)

$$\frac{\partial \psi}{\partial x} = v = xy^2 - 2y - \frac{x^3}{3} \quad (i)$$

$$\frac{\partial \psi}{\partial y} = -u = -\left(\frac{y^3}{3} + 2x - x^2y\right) \quad (ii)$$

Integrating eq. (i) we get

$$\psi = \frac{x^2 y^2}{2} - 2xy - \frac{x^4}{12} + f(y) \quad (iii)$$

Differentiating eq. (iii) with respect to y , we get

$$\frac{\partial \psi}{\partial y} = x^2 y - 2x + f'(y) \quad (iv)$$

Equating the values of $\left(\frac{\partial \psi}{\partial y}\right)$ from eqs. (ii) and (iv), we get,

$$\left(-\frac{y^3}{3} - 2x + x^2 y\right) = x^2 y - 2x + f'(y)$$

$$f'(y) = \frac{-y^4}{12} + C$$



where, C is a numerical constant of integration. Therefore,

$$\psi = \frac{x^2 y^2}{2} - 2xy - \frac{x^4}{12} - \frac{y^4}{12} + C$$

Since C is a numerical constant, it may also be considered as zero.

The velocity field is described as $\vec{V} = Ax^2\hat{i} - 2Axy\hat{j}$.

Is the flow compressible or incompressible? . Find the acceleration of a fluid particle, radius of curvature of the streamline, vorticity at $(x, y) = (1, 3)$.

For two-dimensional incompressible flow, the continuity equation becomes

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

For the given flow,

$$\frac{\partial u}{\partial x} = 2Ax$$

$$\frac{\partial v}{\partial y} = -2Ax$$

$$\text{Thus, } \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

The acceleration of a fluid particle can be obtained as follows:

$$\vec{a} = a_x \hat{i} + a_y \hat{j} = a_s \hat{e}_s + a_n \hat{e}_n \quad (\text{where } s \text{ and } n \text{ are streamline and cross-stream line coordinates})$$

$$a_x = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = (Ax^2)(2Ax) - (2Axy)(0) = 2A^2x^3$$
$$a_y = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = Ax^2(-2Ay) + (-2Axy)(-2Ax) = 2A^2x^2y$$

The acceleration of the fluid particle at (x, y) is

$$\vec{a} = a_x \hat{i} + a_y \hat{j} = 2A^2x^3 \hat{i} + 2A^2x^2y \hat{j}$$

The acceleration of the fluid particle at $(1, 3)$ is

$$\vec{a} = 2A^2(1^3) \hat{i} + 2A^2(1^2)(3) \hat{j} = 2A^2 \hat{i} + 6A^2 \hat{j}$$

The velocity of the fluid particle at $(1, 3)$ is

$$\vec{V} = A(1^2) \hat{i} - 2A(1)(3) \hat{j} = A \hat{i} - 6A \hat{j}$$

The unit vector tangent to the streamline is

$$\hat{\mathbf{e}}_s = \frac{\vec{V}}{|\vec{V}|} = \frac{A\hat{i} - 6A\hat{j}}{\sqrt{(A)^2 + (6A)^2}} = 0.164\hat{i} - 0.986\hat{j}$$

The unit vector normal to the streamline is

$$\hat{\mathbf{e}}_n = \hat{\mathbf{e}}_s \times \hat{\mathbf{k}} = (0.164\hat{i} - 0.986\hat{j}) \times \hat{\mathbf{k}} = -0.986\hat{i} - 0.164\hat{j}$$

The normal component of acceleration is

$$a_n = \vec{a} \cdot \hat{\mathbf{e}}_n = (2A^2\hat{i} + 6A^2\hat{j}) \cdot (-0.986\hat{i} - 0.164\hat{j}) = -2.956A^2$$

The radius of curvature can be found as follows:

$$a_n = -\frac{V^2}{r}$$
$$r = -\frac{V^2}{a_n} = \frac{37A^2}{2.956A^2} = 12.52$$

The velocity field in a fluid medium is given by $\vec{V} = 3xy^2\vec{i} + 2xy\vec{j} + (2zy + 3t)\vec{k}$

Find the magnitudes and directions of (i) translational velocity, (ii) rotational velocity and (iii) the vorticity of a fluid element at (1, 2, 1) and at time $t=3$.

(i) Translational velocity vector at (1,2,1) and at $t = 3$ can be written as,

$$\vec{V} = 3(1)(4)\vec{i} + 2(1)(2)\vec{j} + (2.1.2 + 3.3)\vec{k} = 12\vec{i} + 4\vec{j} + 13\vec{k}$$

Hence

x component of translational velocity

$$u = 12 \text{ units}$$

y component of translational velocity

$$v = 4 \text{ units}$$

z component of translational velocity

$$w = 13 \text{ units}$$

(ii) Rotational velocity vector is found as

$$\begin{aligned}\vec{\omega} &= \frac{1}{2} (\nabla \times \vec{V}) = \frac{1}{2} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} \\ &= \frac{\vec{i}}{2} \left\{ \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right\} + \frac{\vec{j}}{2} \left\{ \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right\} + \frac{\vec{k}}{2} \left\{ \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right\} \\ &= \vec{i} \frac{1}{2} \left[\frac{\partial}{\partial y} (2zy + 3t) - \frac{\partial}{\partial z} (2xy) \right]\end{aligned}$$

$$\begin{aligned}
& + \vec{j} \frac{1}{2} \left[\frac{\partial}{\partial z} (3xy^2) - \frac{\partial}{\partial x} (2zy + 3t) \right] \\
& + \vec{k} \frac{1}{2} \left[\frac{\partial}{\partial x} (2xy) - \frac{\partial}{\partial y} (3xy^2) \right] \\
& = z\vec{i} + (y - 3xy)\vec{k}
\end{aligned}$$

at (1,2,1) and $t = 3$,

$$\vec{\omega} = \vec{i} - 4\vec{k}$$

(iii) The vorticity $\vec{\Omega} = 2\vec{\omega}$